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Student Number

2021 Glenwood High School
Year 12 – Trial HSC Examination

Mathematics Extension 1

**General
Instructions**

- * Reading Time – 10 minutes
- * Working time – 2 hours
- * Write using black pen
- * NESA approved calculators may be used
- * A reference sheet is provided
- * For questions in Section II, show relevant mathematical reasoning and/or calculations

**Total marks:
70**

Section I – 10 marks (pages 2 – 5)

- * Attempt Questions 1-10
- * Allow about 15 minutes for this section

Section II – 60 marks (pages 6 – 11)

- * Attempt Questions 11 – 14
- * Allow about 1 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10.

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 10.

1. What is the value of $\cos(75^\circ)$?

A. $\frac{\sqrt{3}-1}{2\sqrt{2}}$

B. $\frac{\sqrt{6}-\sqrt{2}}{2}$

C. $\frac{\sqrt{3}+1}{2\sqrt{2}}$

D. $\frac{\sqrt{6}+\sqrt{2}}{2}$

2. The vectors $4\hat{i} - \hat{j}$ and $3\hat{i} + x\hat{j}$ are perpendicular. What is the value of x ?

A. -12

B. -1

C. 3

D. 12

3. The probability of Hannah scoring a goal in a netball game is 0.45. What is the least number of games that must be played to ensure the probability of scoring a goal at least once is more than 0.9?

A. 2

B. 3

C. 4

D. 5

4. What is the domain and range of the function $f(x) = 2 \sin^{-1} 3x$?

- A. $-\frac{1}{3} \leq x \leq \frac{1}{3}, -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
- B. $-\frac{1}{3} \leq x \leq \frac{1}{3}, -\pi \leq y \leq \pi$
- C. $-3 \leq x \leq 3, -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
- D. $-3 \leq x \leq 3, -2 \leq y \leq 2$

5. A sock drawer has 8 black socks and 14 white socks.

What is the minimum number of times someone needs to remove socks to ensure that two different coloured socks are selected?

- A. 2
- B. 14
- C. 15
- D. 22

6. The variance of a Bernoulli distribution is $\frac{3}{16}$.

What is a possible value for the mean of the distribution?

- A. $\frac{3}{16}$
- B. $\frac{5}{16}$
- C. $\frac{3}{4}$
- D. $\frac{5}{4}$

7. A particle is initially positioned at $x = -2$. Its motion has velocity of $v = \frac{1}{t+e}$.

Which of the following will be true when the particle reaches the origin?

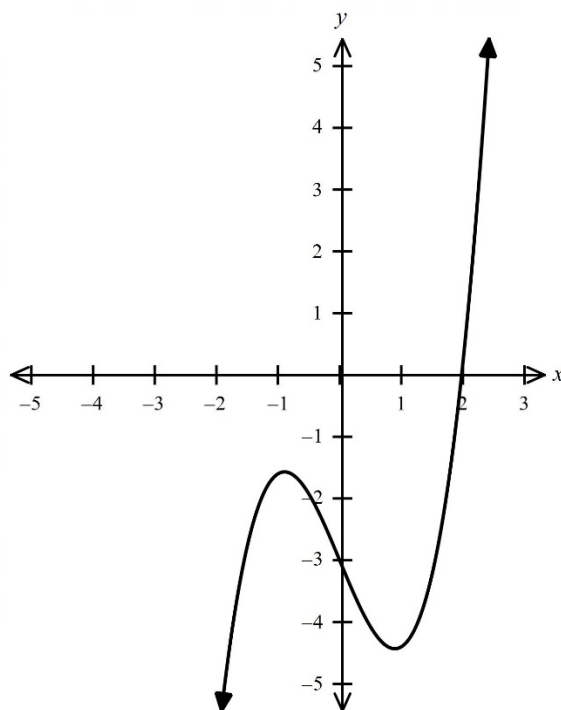
- A. $t = 2, v = \frac{1}{2+e}$
- B. $t = e, \dot{x} = \frac{1}{2e}$
- C. $t = e^2, v = \frac{1}{e^2+e}$
- D. $t = e^3 - e, v = e^{-3}$

8. It is known that $\sin x = \frac{1}{3}$, where $\frac{\pi}{2} < x < \pi$.

What is the value of $\cos 2x$?

- A. $-\frac{7}{9}$
- B. $\frac{7}{9}$
- C. $\frac{2}{3}$
- D. $\frac{-4\sqrt{2}}{3}$

9. Consider the graph of $y = f(x)$ shown here:



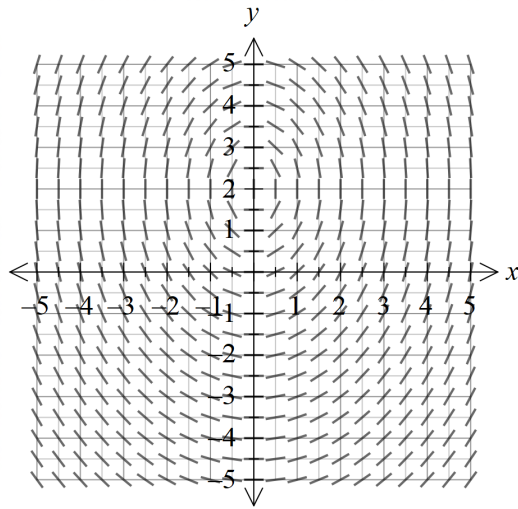
Which one of the following would have 2 more roots than $f(x)$?

- A. $y = -2 \times f(x)$
- B. $y = f(x) + 3$
- C. $y = f^{-1}(x)$
- D. $y = f(x + 3)$

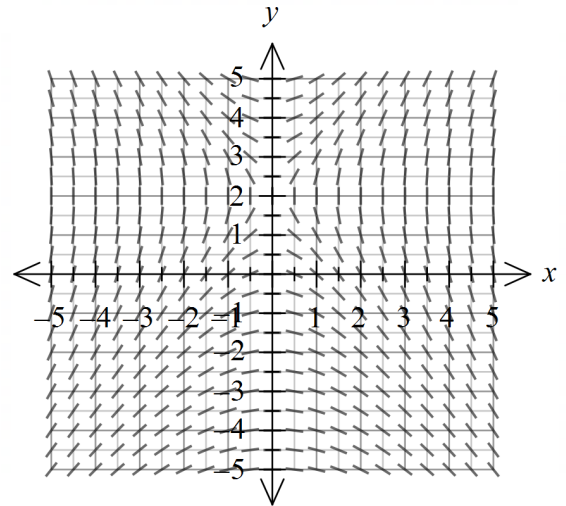
10. Which of the following best represents the direction field for the differential equation

$$\frac{dy}{dx} = \frac{-2x}{y-2} ?$$

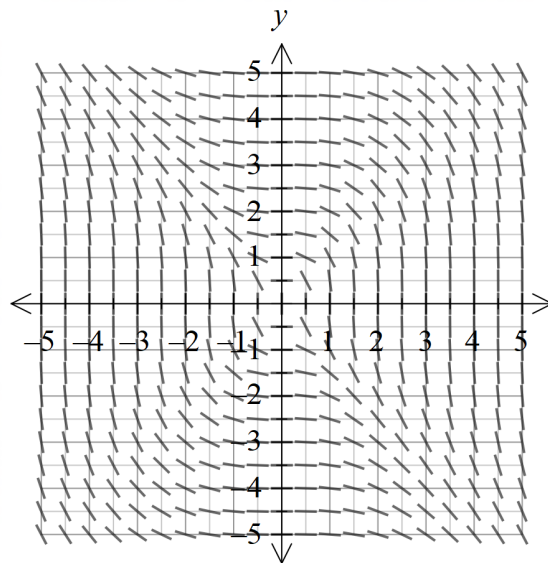
A.



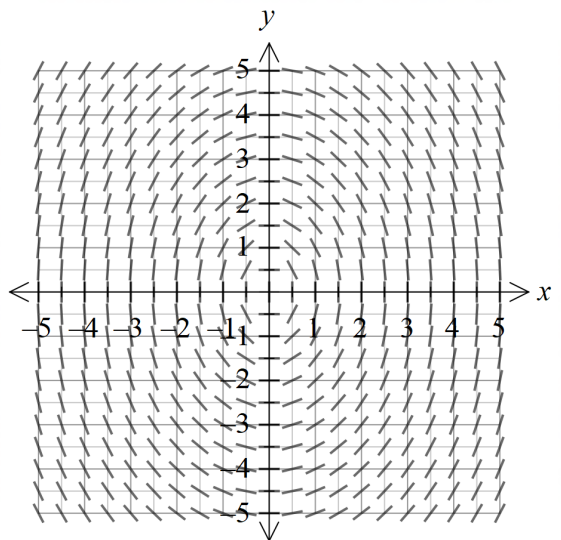
B.



C.



D.



Section II

60 marks

Attempt Questions 11 – 14.

Allow about 1 hour and 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

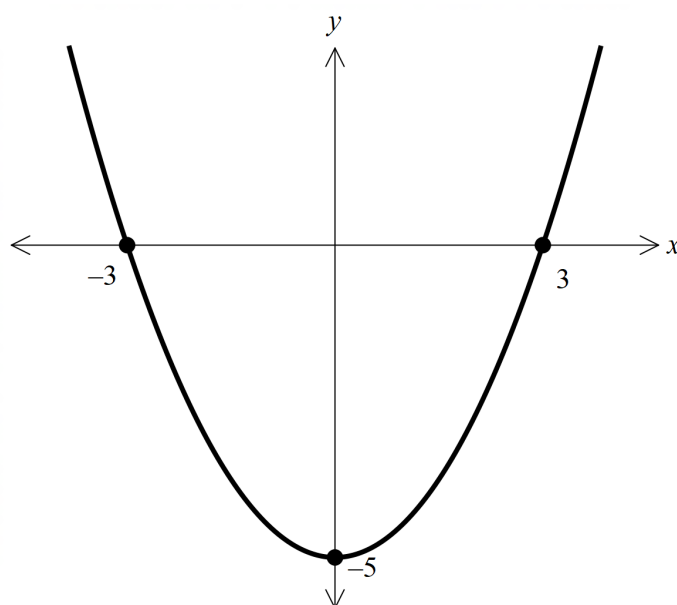
In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (16 marks) Use a new writing booklet.

(a) For what values of x is $\frac{x}{x-1} \geq -2$?

3

(b) The diagram shows the graph of $y = f(x)$.



(i) Sketch a one-third page diagram of $y = f^{-1}(x)$ after making an appropriate restriction on the domain of $f(x)$ so that $y = f^{-1}(x)$ is a one-to-one increasing function. Show intercepts.

1

(ii) Sketch $y = \frac{1}{f^{-1}(x)}$ on the same set of axes, showing all necessary features

2

Question 11 continues on page 7

Question 11 continued

- (c) Let $P(x) = x^3 + 5x^2 + x - 15$. **1**
- (i) Show that $P(-3) = 0$. **1**
- (ii) Hence, express the polynomial $P(x)$ as $A(x)B(x)$, where $B(x)$ is a quadratic polynomial. **2**
- (d) Triangle ABC has vertices $A(-4, 4)$, $B(1, 7)$ and $C(10, 0)$. **2**
- (i) Give $\overrightarrow{AB}$ and $\overrightarrow{AC}$ in column vector form. **2**
- (ii) Find $\angle BAC$ to the nearest minute and hence find the area of the triangle. **2**
- (e) Show that $\int_0^{\frac{\pi}{2}} \sin 7x \sin 3x dx = 0$ **3**

End of Question 11

Please turn over

Question 12 (15 marks) Use a new writing booklet.

- (a) Use the principle of mathematical induction to show that for all integers $n \geq 1$ 3

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2}{4} (n+1)^2.$$

- (b) A particle is projected from level ground with an initial velocity of

$5\mathbf{i} + 12\mathbf{j} \text{ ms}^{-1}$. Use $g = 9.8 \text{ ms}^{-2}$.

- (i) Find the initial speed and angle of projection of the particle. 2

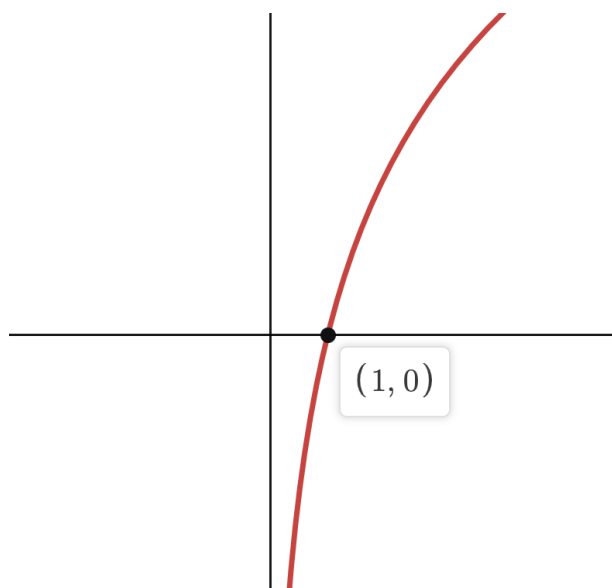
- (ii) Show that the displacement vector $\underline{d} = 5t\underline{i} + (12t - 4.9t^2)\underline{j}$ and 3

hence find the time of flight of the particle to one decimal place.

- (iii) Find the horizontal distance travelled by the particle. 1

- (iv) Find the maximum height reached by the particle. 2

- (c) The graph of $y = 4\ln x$ is given below. The part of the graph between $x = 1$ and $x = e$ is rotated about the y -axis. Find the exact volume of the solid formed. 4

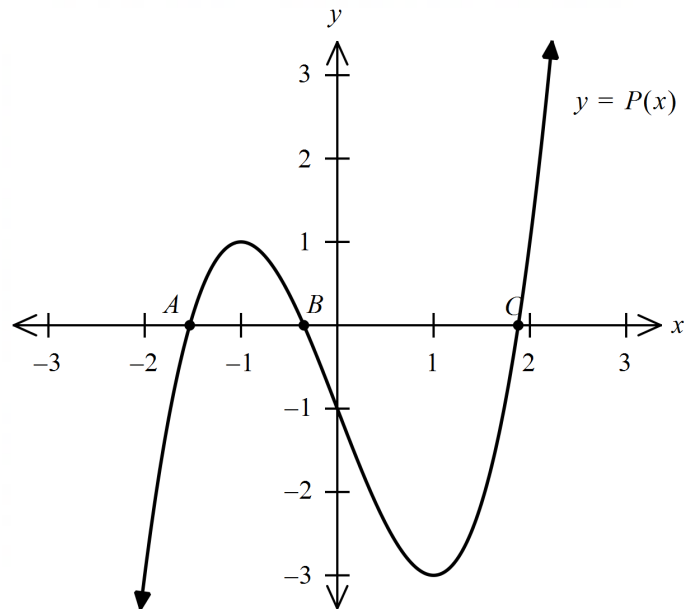


End of Question 12

Question 13 (14 marks) Use a new writing booklet.

(a) (i) Show that $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$. 2

(ii) The graph of the polynomial $y = P(x) = x^3 - 3x - 1$ is shown below. 4



Let $x = 2\sin\theta$ and by using part (i), find the roots of $P(x)$: A , B and C , correct to 2 decimal places.

(b) Use the substitution $u = x^3 + 2$ to find the exact value of $\int_0^3 2x^2 \sqrt{x^3 + 2} dx$ 2

(c) Using the substitution $t = \tan \frac{x}{2}$, solve the equation 3

$$2 \sin x - 5 \cos x - 5 = 0 \text{ for } 0 \leq x \leq \pi.$$

(d) Water is poured into a container at a rate of $8 \text{ cm}^3 \text{ s}^{-1}$ and the volume of the water in the container is given by 3

$$V = \frac{3}{2}(h^2 + 8h) \quad \text{where } h \text{ cm is the depth of the water.}$$

Find the rate of change of the depth of the water when $V = 72 \text{ cm}^3$.

End of Question 13

Question 14 (15 marks) Use a new writing booklet.

- (a) Solve the differential equation $\frac{dy}{dx} = \frac{e^x}{2y}$, given the initial condition $y(2) = e$. 3

- (b) A store has reduced the prices on two of their sale items as there had been a number of faults reported by customers.

One of the items is a movie DVD. The store has 3 of these left in stock and it is known that there is a 2.5% probability that each is faulty.

The other item is a music CD with each CD having a 10.8% probability of being faulty. The store has 5 of these left to sell.

A customer chooses to purchase 2 of the movie DVDs and 3 of the music CDs as gifts.

- (i) What is the probability that none of these are faulty? 2

- (ii) What is the probability that at least one of the items is faulty? 1

- (c) The equation $y = 7 + \cos\left(\frac{4\pi}{25}t\right) - \sin\left(\frac{4\pi}{25}t\right)$ gives the water depth y metres at the wharf, t hours after high tide which occurred at 2 am that morning.

- (i) State the amplitude. 1

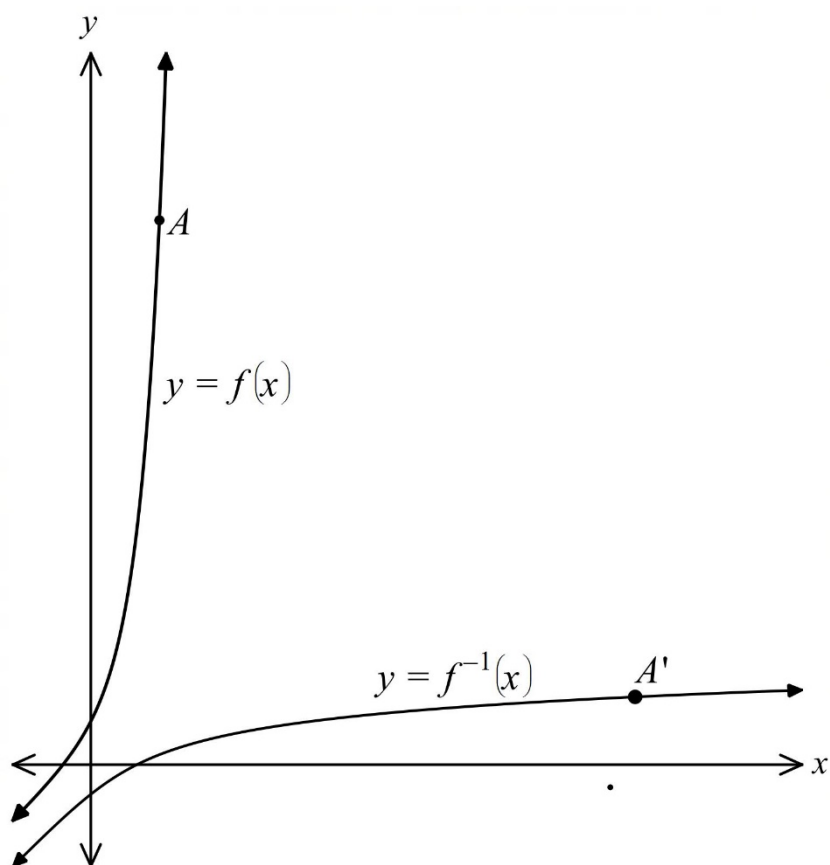
- (ii) An overhead power cable obstructs a ship's exit from the wharf. The ship will only be able to exit the wharf if the depth of the water is 8 metres or less. 3

Find the *latest* possible time before 4 pm that day when the ship can leave the wharf.

Question 14 continues on page 11

Question 14 continued

- (d) Consider the graphs of $y = f(x)$ and $y = f^{-1}(x)$ where $f(x) = 1 + x + e^x$ shown below.



The point A lies on $y = f(x)$ and has an x -coordinate of 3. The point A' is the image of point A on $y = f^{-1}(x)$.

- (i) Find the exact coordinates of A' . 1
- (ii) It is given that the gradient of the tangent to $y = f(x)$ at A and the gradient to the tangent to $y = f^{-1}(x)$ at A' are reciprocal of each other. Show that the equation of the tangent to $y = f^{-1}(x)$ at A' is $y = \frac{x - 1 + 2e^3}{1 + e^3}$. 2
- (iii) Find the coordinates of the point where the tangent to $y = f(x)$ at A meets the tangent from part (ii).

End of Examination

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Section 1

$$\begin{aligned} 01) \cos 75 &= \cos(30 + 45) \\ &= \cos 30 \cos 45 - \sin 30 \sin 45 \\ &= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} - \frac{1}{2} \times \frac{1}{\sqrt{2}} = \frac{\sqrt{3}-1}{2\sqrt{2}} \end{aligned}$$

(A)

$$02) \frac{4x^3 + -1x^2}{\sqrt{4^2+3^2} \times \sqrt{9x^2}} = 0$$

$$\frac{12-x}{5 \times 3x} = 0$$

(D)

$$12-x = 0$$

$$x = 12$$

$$03) 1 - (0.55)^n > 0.9$$

$$-0.55^n > -0.1$$

$$0.55^n < 0.1$$

$$n \ln 0.55 < \ln 0.1$$

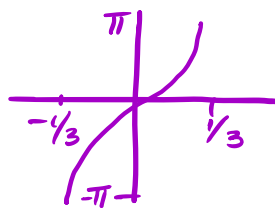
$$n > \frac{\ln 0.1}{\ln 0.55}$$

$$n > 3.85$$

$$n = 4$$

(C)

04)



$$-\frac{1}{3} \leq x \leq \frac{1}{3}$$

$$-\pi \leq y \leq \pi$$

(B)

$$05) 15 \quad (C)$$

$$06) p(1-p) = \frac{3}{16}$$

by trial x error

$$\frac{3}{16} \times \frac{13}{16} \neq \frac{3}{16}$$

$$\frac{5}{16} \times \frac{11}{16} \neq \frac{3}{16}$$

$$\frac{3}{4} \times \frac{1}{4} = \frac{3}{16} \quad \checkmark$$

$$\text{At } t=0, \quad x=-2$$

$$v = \frac{1}{t+e}$$

$$87) \quad x = \ln(t+e) + C$$

$$-2 = \ln e + C$$

$$-3 = C$$

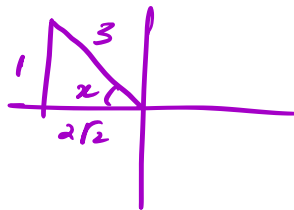
$$\therefore x = \ln(t+e) - 3$$

$$\text{At } x=0, \quad \ln(t+e) = 3$$

$$e^3 = t+e$$

$$e^3 - e = t$$

$$88) \quad \sin x = \frac{1}{3}$$



$$\cos 2x = \cos^2 x - \sin^2 x$$

$$= \left(\frac{2\sqrt{2}}{3}\right)^2 - \left(\frac{1}{3}\right)^2$$

$$= \frac{8}{9} - \frac{1}{9} = \frac{7}{9}$$

$$89) \quad B$$

$$916) \quad \frac{dy}{dx} = \frac{-2x}{y-2}$$

$$\text{when } y=0 \quad \text{as } x > 0 \quad \underline{dy} > 0$$

Section 2

Q1) a) $\frac{x}{x-1} \geq -2$

$$x(x-1) \geq -2(x-1)^2$$

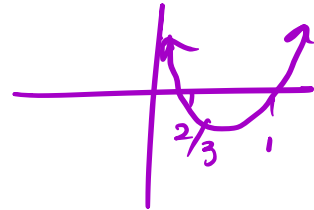
$$x(x-1) + 2(x-1)^2 \geq 0$$

$$(x-1)[x + 2(x-1)] \geq 0 \quad \checkmark$$

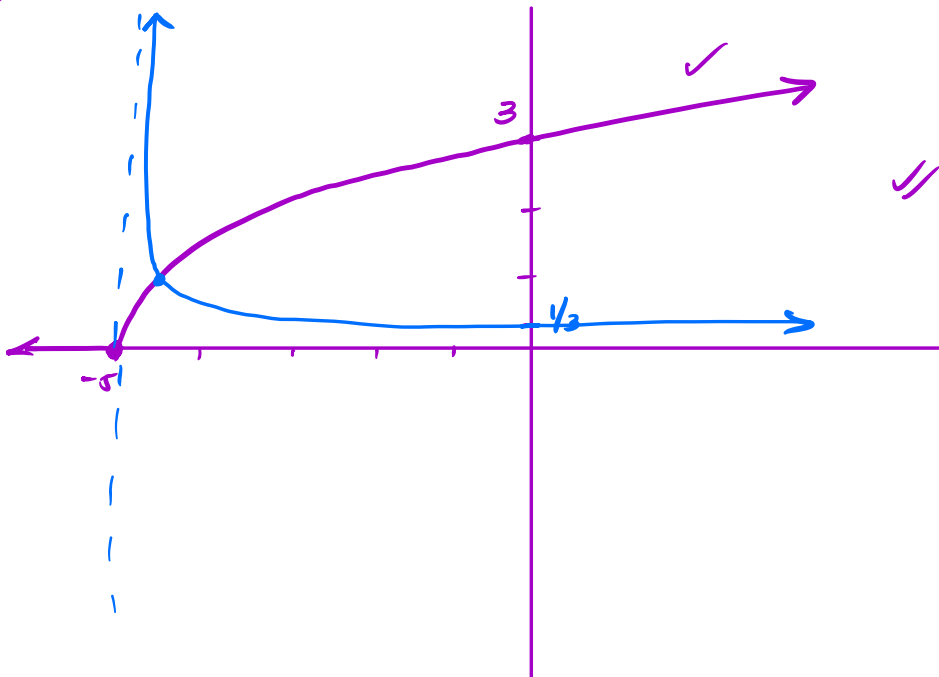
$$(x-1)[x + 2x - 2] \geq 0$$

$$(x-1)(3x-2) \geq 0$$

$$x \leq \frac{2}{3} \quad x > 1 \quad \checkmark$$



b). (1) $x(u)$



c) (i) $P(x) = x^3 + 5x^2 + x - 15$

$$P(-3) = (-3)^3 + 5(-3)^2 + (-3) - 15$$

$$= -27 + 45 - 3 - 15$$

$$= 0 \quad \checkmark$$

(ii) $(x+3)$ is a factor since $P(-3) = 0$

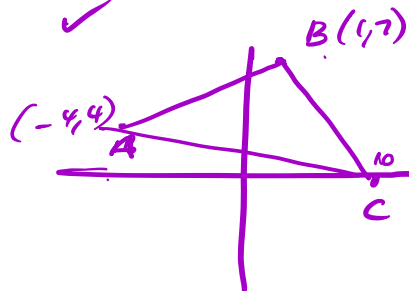
$$\begin{array}{r} x^2 + 2x - 5 \\ x+3 \overline{) x^3 + 5x^2 + x - 15} \\ \underline{x^3 + 3x^2} \\ 2x^2 + x \\ \underline{2x^2 + 6x} \\ -5x - 15 \\ \underline{-5x - 15} \\ 0 \end{array} \quad \checkmark \checkmark$$

$$\therefore P(x) = (x+3)(x^2 + 2x - 5)$$

d) (i) $A = (-4, 4) \quad B = (1, 7) \quad C = (10, 0)$

$$\vec{AB} = \begin{pmatrix} 1+4 \\ 7-4 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} \quad \checkmark$$

$$\vec{AC} = \begin{pmatrix} 10+4 \\ 0-4 \end{pmatrix} = \begin{pmatrix} 14 \\ -4 \end{pmatrix}$$



$$\begin{aligned} \text{(ii)} \quad \theta &= \cos^{-1} \left[\frac{5 \times 14 - 3 \times 4}{\sqrt{5^2 + 3^2} \times \sqrt{14^2 + (-4)^2}} \right] \\ &= 46^\circ 55' \quad \checkmark \end{aligned}$$

$$\begin{aligned} A &= \frac{1}{2} \times \sqrt{5^2 + 3^2} \times \sqrt{14^2 + 16} \times \sin 46^\circ 55' \\ &= 31.004 \approx 31 \text{ u}^2 \quad \checkmark \end{aligned}$$

$$\begin{aligned}
 \text{e). } & \int_0^{\pi/2} \sin 7x \sin 3x \, dx \\
 &= \frac{1}{2} \int_0^{\pi/2} (\cos 4x - \cos 10x) \, dx \quad \checkmark \\
 &= \frac{1}{2} \left[\frac{\sin 4x}{4} - \frac{\sin 10x}{10} \right]_0^{\pi/2} \quad \checkmark \\
 &= \frac{1}{2} \left[\frac{\sin 2\pi}{4} - \frac{\sin 5\pi}{10} - \frac{\sin 0}{4} + \frac{\sin 0}{10} \right] \\
 &= \frac{1}{2} [0 - 0 - 0 + 0] = 0 \quad \checkmark
 \end{aligned}$$

Q12)

a). Prove the result is true for $n=1$

$$\text{LHS} = 1^3 = 1$$

$$\text{RHS} = \frac{1}{4} (1+1)^2 = \frac{1}{4} \times 4 = 1 = \text{LHS}$$

$\therefore$ true for $n=1$

Assume it's true for $n=k$

$$1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{k^2}{4} (k+1)^2$$

Prove it's true for $n=k+1$

$$1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = \frac{(k+1)^2}{4} (k+2)^2$$

LHS

$$\underbrace{1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3}_{S(k)}$$

$$\frac{k^2}{4} (k+1)^2 + (k+1)^3$$

$$= (k+1)^2 \left[\frac{k^2}{4} + k+1 \right]$$

$$= (k+1)^2 \left[\frac{k^2 + 4k + 4}{4} \right]$$

$$= (k+1)^2 \left(\frac{k+2}{4} \right)^2$$

$$= \frac{(k+1)^2}{4} (k+2)^2$$

$$= \text{RHS}$$

$\therefore$ true for $n=k+1$, if the result is true for $n=k$
Then it's true for all $n \geq 1$

$$b) \quad \underline{\underline{v}} = 5i + 12j$$

$$|\underline{\underline{v}}| = \sqrt{5^2 + 12^2} = \sqrt{169}$$

$$\tan \theta = \frac{12}{5}$$

$$\theta = \tan^{-1} \left(\frac{12}{5} \right)$$

$$= 67^\circ 23'$$

$$(ii) \quad \vec{a} = -g\hat{j}$$

$$\vec{v} = -gt\hat{j} + C$$

$$\text{At } t=0, \quad \vec{v} = 5\hat{i} + 12\hat{j}$$

$$5\hat{i} + 12\hat{j} = C$$

$$\vec{v} = -gt\hat{j} + 5\hat{i} + 12\hat{j}$$

$$\vec{v} = 5\hat{i} + (12 - gt)\hat{j}$$

$$\vec{d} = 5t\hat{i} + \left(12t - \frac{gt^2}{2}\right)\hat{j} + C$$

$$\text{At } t=0 \quad \vec{d} = 0\hat{i} + 0\hat{j}$$

$$\therefore C = 0$$

$$\vec{d} = 5t\hat{i} + \left(12t - \frac{gt^2}{2}\right)\hat{j}$$

$$\text{for time of flight, } 12t - \frac{gt^2}{2} = 0$$

$$12t - \frac{9.8t^2}{2} = 0$$

$$t(12 - 4.9t) = 0$$

$$t = 0, \quad t = \frac{120}{49}$$

$$(iii) \quad \text{At } t = \frac{120}{49}, \quad x = 5 \times \frac{120}{49} = \frac{600}{49} = 12.24$$

$$(iv) \quad \text{For max height, } 12 - gt = 0$$

$$12 - 9.8t = 0$$

$$9.8t = 12$$

$$t = \frac{12}{9.8}$$

$$\text{At } t = \frac{12}{9.8}, \quad y = 12\left(\frac{12}{9.8}\right) - \frac{9.8}{2} \times \left(\frac{12}{9.8}\right)^2$$

$$= \frac{360}{49} = 7.34$$

c). $y = 4 \ln x$

when $x=1$, $y = 4 \ln 1 = 0$
 $x=e$, $y = 4 \ln e = 4$

$$\frac{y}{4} = \ln x$$

$$x = e^{y/4}$$

$$V = \pi \int_0^4 (e^{y/4})^2 dy = \pi \int_0^4 e^{y/2} dy$$

(u) $V = \pi \left[\frac{e^{y/2}}{1/2} \right]_0^4$

$$= 2\pi [e^{y/2}]_0^4$$

$$= 2\pi [e^2 - e^0] = 2\pi [e^2 - 1] u^3$$

Q13) a) i) $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$

$$\sin(2\theta + \theta) = \sin 2\theta \cos \theta + \cos 2\theta \sin \theta$$

$$= 2 \sin \theta \cos \theta \cos \theta + (1 - 2 \sin^2 \theta) \sin \theta$$

$$= 2 \sin \theta (1 - \sin^2 \theta) + (1 - 2 \sin^2 \theta) \sin \theta$$

$$= 2 \sin \theta - 2 \sin^3 \theta + \sin \theta - 2 \sin^3 \theta$$

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

$$\begin{aligned} (11). \quad P(x) &= (2\sin\theta)^3 - 3(2\sin\theta) - 1 \\ &= 8\sin^3\theta - 6\sin\theta - 1 \\ &= -2(4\sin^3\theta + 3\sin\theta) - 1 \end{aligned}$$

from part (a)

$$-4\sin^3\theta + 3\sin\theta = \sin 3\theta$$

$$\therefore P(x) = -2(\sin 3\theta) - 1$$

for roots

$$-2(\sin 3\theta) - 1 = 0$$

$$\sin 3\theta = -\frac{1}{2}$$

$$\text{base angle} = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

$$\therefore 3\theta = -\frac{\pi}{6}, -\left(\pi - \frac{\pi}{6}\right), \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$= -\frac{\pi}{6}, -\frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\theta = -\frac{\pi}{18}, -\frac{5\pi}{18}, \frac{7\pi}{18}, \frac{11\pi}{18}$$

$$\begin{aligned} \therefore x &= 2\sin\left(-\frac{\pi}{18}\right), 2\sin\left(-\frac{5\pi}{18}\right), 2\sin\frac{7\pi}{18} \\ &\quad 2\sin\frac{11\pi}{18} \dots \end{aligned}$$

$$= -0.35, -1.53, 1.88, 1.88$$

from the location of A, B, C from graph given

$$x = -0.35, -1.53, 1.88$$

$$b) \int_0^3 2x^2 \sqrt{x^3+2} \, dx$$

$$u = x^3 + 2$$

$$\frac{du}{dx} = 3x^2$$

$$\frac{2}{3} \int_0^3 3x^2 \sqrt{x^3+2} \, dx$$

$$du = 3x^2 dx$$

$$\text{when } x=0, \quad u=2$$

$$x=3, \quad u=29$$

$$\frac{2}{3} \int_2^{29} \sqrt{u} \, du$$

$$\frac{2}{3} \left[\frac{u^{3/2}}{3/2} \right]_2^{29}$$

$$\frac{2}{3} \times \frac{2}{3} \left[29^{3/2} - 2^{3/2} \right]$$

$$= \frac{4}{9} \left(\sqrt{29^3} - \sqrt{2^3} \right)$$

$$= \frac{4}{9} \left(29\sqrt{29} - 2\sqrt{2} \right)$$

$$c). \quad t = \tan \frac{x}{2}$$

$$2\sin\theta - 5\cos\theta - 5 = 0$$

$$2\left(\frac{2t}{1+t^2}\right) - 5\left(\frac{1-t^2}{1+t^2}\right) - 5 = 0$$

$$4t - 5 + 5t^2 - 5(1+t^2) = 0$$

$$4t - 5 + 5t^2 - 5 - 5t^2 = 0$$

$$4t - 10 = 0$$

$$t = 10/4 = 5/2 \quad 0 \leq x \leq \pi$$

$$\tan x/2 = 5/2 = 2.5 \quad 0 \leq x/2 \leq \pi/2$$

$$x/2 = \tan^{-1}(2.5)$$

$$x = 2\tan^{-1}(2.5)$$

$$\text{Test } x = \pi$$

$$\underline{\text{LHS}} \quad 2 \sin \pi - 5 \cos \pi - 5 = 0 = \text{RHS}$$

$\therefore$ Solutions are $2\tan^{-1}(2.5), \pi$

$$d). \quad \frac{dV}{dt} = 8$$

$$V = \frac{3}{2}(h^2 + 8h)$$

$$\frac{dV}{dh} = \frac{3}{2}(2h + 8)$$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$= \frac{2}{3(2h+8)} \times 8 = \frac{16}{3(2h+8)}$$

$$\text{when } V = 72$$

$$72 = \frac{3}{2}(h^2 + 8h)$$

$$\frac{72 \times 2}{3} = h^2 + 8h$$

$$48 = h^2 + 8h$$

$$h^2 + 8h - 48 = 0$$

$$(h-4)(h+12) = 0$$

$$h = 4, -12 \quad \text{Since } h > 0, \quad h = 4$$

$$\therefore \frac{dh}{dt} = \frac{16}{3(2 \times 4 + 8)} = \frac{1}{3}$$

$$14 \text{ a) } \frac{dy}{dx} = \frac{e^x}{2y} \quad y(2) = e$$

$$\int 2y \, dy = \int e^x \, dx$$

$$\frac{2y^2}{2} = e^x + C$$

$$e^2 = e^2 + C$$

$$C = 0$$

$$\therefore y^2 = e^x$$

$$y = \pm \sqrt{e^x}$$

checking initial condition

$$y = \sqrt{e^x}$$

$$b) \quad 1 \binom{3}{2} (0.025)^1 (0.975)^2 \times \binom{5}{3} (0.108)^2 (0.892)^3$$
$$= 0.59\%$$

$$(u) \quad 1 - 0.59\% = 99.41\%$$

$$c). \quad y = 7 + \cos\left(\frac{4\pi}{25}t\right) - \sin\left(\frac{4\pi}{25}t\right)$$

$$\cos\left(\frac{4\pi}{25}t\right) - \sin\left(\frac{4\pi}{25}t\right) = \sqrt{2} \cos\left(\frac{4\pi}{25}t + \frac{\pi}{4}\right)$$

$$R = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\alpha = \tan^{-1} 1/1 = \pi/4$$

$$\text{Amplitude} = \sqrt{2}$$

$$7 + \cos\left(\frac{4\pi}{25}t\right) - \sin\left(\frac{4\pi}{25}t\right) = 8$$

$$7 + \sqrt{2} \cos\left(\frac{4\pi}{25}t + \frac{\pi}{4}\right) = 8$$

$$T_2 \cos \left(\frac{4\pi}{25}t + \frac{\pi}{4} \right) = 1$$

$$\cos \left(\frac{4\pi}{25}t + \frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}$$

$$\text{Base angle} = \pi/4$$

$$\frac{4\pi}{25}t + \frac{\pi}{4} = \frac{\pi}{4}, \frac{2\pi - \pi}{4}, \frac{2\pi + \pi}{4}, \frac{2\pi + 2\pi - \pi}{4}$$

$$= \frac{\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \frac{15\pi}{4}$$

$$\frac{4\pi}{25}t = 0, \frac{6\pi}{4}, \frac{8\pi}{4}, \frac{14\pi}{4}$$

$$= 0, \frac{8\pi}{2}, 2\pi, \frac{7\pi}{2}$$

$$t = 0, 9.375, 12.5, 21.875$$

$t = 12.5$ is the latent before 4pm that day which is 2:30 pm

$$d) y = 1 + x + e^x$$

$$\text{At } x = 3, y = 1 + 3 + e^3 = 4 + e^3$$

$$\therefore A' \text{ is } (4 + e^3, 3)$$

$$y' = 1 + e^x$$

$$\text{At } x = 3$$

$$y' = 1 + e^3$$

$$\text{Gradient @ } A' = \frac{1}{1+e^3}$$

Equation of tangent,

$$y - 3 = \frac{1}{1+e^3} [x - (4 + e^3)]$$

$$y - 3 = \frac{1}{1+e^3} (x - 4 - e^3)$$

$$y = \frac{x - 4 - e^3}{1+e^3} + 3$$

$$y = \frac{x - 4 - e^3 + 3(1+e^3)}{1+e^3}$$

$$y = \frac{x - 4 - e^3 + 3 + 3e^3}{1+e^3}$$

$$y = \frac{x + 2e^3 - 1}{1+e^3}$$

(iii) The tangents meet on $y = x$

$$\frac{x + 2e^3 - 1}{1+e^3} = x$$

$$x + 2e^3 - 1 = x(1+e^3)$$

$$\cancel{x} + 2e^3 - 1 = \cancel{x} + xe^3$$

$$x = \underline{2e^3 - 1}$$

$$\frac{\quad}{e^3}$$

$$\text{At } x = \frac{2e^3 - 1}{e^3}$$

$$y = \frac{\frac{2e^3 - 1}{e^3} + 2e^3 - 1}{1 + e^3}$$

$$= \frac{2e^3 - 1 + e^3(2e^3 - 1)}{e^3(1 + e^3)}$$

$$= \frac{(2e^3 - 1)(\cancel{1 + e^3})}{e^3(\cancel{1 + e^3})} = \frac{2e^3 - 1}{e^3}$$

$$\therefore \left(\frac{2e^3 - 1}{e^3}, \frac{2e^3 - 1}{e^3} \right)$$